

Lab02. EC2 and Cloud9.

Cloud Programming (W04IST–SI4527G)

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1 Learning goals

1. Create Cloud9 environment.
2. Configure passwordless access to GitHub.
3. Working with GitHub repositories in Cloud9.

2 Prerequisites

None.

3 Exercises

3.1 Create your AWS Cloud9 environment

In this step, you use the AWS Cloud9 console to create and then open an AWS Cloud9 development environment.

1. Choose the **Create environment** button.
2. On the Create environment page, for Name, enter a name for your environment.
 - For **Description**, enter something about your environment. For this tutorial, use “Cloud development IDE”.
 - For **Environment type**, choose **New EC2 instance** to create an Amazon EC2 environment.
 - On the **New EC2 instance**, keep the default choice. This option might have less RAM and fewer vCPUs. However, this amount of memory is sufficient for this tutorial.
 - For **Platform**, choose the type of Amazon EC2 instance of your choice: Amazon Linux 2023, Amazon Linux 2 or Ubuntu 22.04 LTS. AWS Cloud9 creates the instance and then connects the environment to it.
 - Choose a time period for **Timeout**. This option determines how long AWS Cloud9 is inactive before auto-hibernating. When all web browser instances that are connected to the IDE for the environment are closed, AWS Cloud9 waits the amount of time specified and then shuts down the Amazon EC2 instance for the environment.

- **IMPORTANT:** On the Network settings panel, choose **Secure Shell (SSH)** – this method accesses the environment using SSH and requires open inbound ports.
3. Choose **Create** to create your environment, and then you're redirected to the home page. If the account is successfully created, a green flash bar appears at the top of the AWS Cloud9 console. You can select the new environment and choose **Open** in Cloud9 to launch the IDE.
 4. Look around the Cloud9 environment.

3.2 Configure GitHub passwordless login

In this exercise, you will:

- create a key pair for SSH authentication,
- add private key to SSH agent,
- upload private key to GitHub,
- configure your Git settings,
- modify GitHub repository.

3.2.1 Create a new SSH key pair

1. In Cloud9 bash window (at bottom), type command to create a new key pair:

```
$ ssh-keygen -t ed25519 -C "<YOUR-INITIALS> CloudProg2026"
```

replace `<YOUR-INITIALS>` with your initials, eg.: `"WT CloudProg2026"`. When prompted, enter:

- “Enter file in...”: `/home/ec2-user/.ssh/<YOUR-INITIALS>_cloudprog2026`
- “Enter passphrase”: enter password to protect your private key.

2. Add your private key to SSH agent, to avoid entering password every time:

```
$ eval $(ssh-agent -s)
$ ssh-add ~/.ssh/<YOUR-INITIALS>_cloudprog2026
```

Enter your key password, when prompted.

You can read more about creating SSH keys in [\[1\]](#).

3.2.2 Add public key to GitHub

1. In the browser, login to your GitHub account. Click your avatar (in top right corner) and select “Settings”. On the left, in “Access” section, click “SSH and GPG keys”. Click the button “New SSH key”.
2. In the title type: `<YOUR-INITIALS> Cloud Programming 2026 key`
3. In Cloud9 bash, type: `cat ~/.ssh/<YOUR-INITIALS>_cloudprog2026.pub`
4. Copy displayed line: `ssh-ed25519 AAA...XHW CloudProg2026` , and paste it into “Key” text field on GitHub. Click “Add SSH key”.

You can read more about creating SSH keys in [\[2\]](#).

3.2.3 Configure Git

You must always configure your username and e-mail, in order to be able to save (commit) changes to Git repositories.

1. In Cloud9 bash, execute commands:

```
$ git config --global user.name "<YOUR NAME>"
$ git config --global user.email "<YOUR EMAIL>"
```

You can read more about using Git in [3].

3.2.4 Verification

In this task, you will clone lab repository from GitHub to Cloud9, make some changes locally, commit changes, push changes back to GitHub.

1. On ePortal, open Assignment 1. Login to GitHub. If you open assignment for the first time - select your name from the list. Accept the assignment (a new repository is created for you). Click the repository link, to open it.
2. In GitHub assignment repository, click green button “Code”, select “Local” tab, select “SSH” tab, copy the address (you can click double-square to copy address). It looks like: `git@github.com:...git`.
3. In Cloud9 bash, clone repository with assignment, and enter into the cloned folder with the repository:

```
$ git clone <COPIED-ADDRESS>
$ ls
$ cd clprog-lab-1-<YOUR-NAME>
$ git status
```

4. You should see:

```
On branch main
Your branch is up to date with 'origin/main'.

nothing to commit, working tree clean
```

5. In Cloud9, on left sidebar select folder icon (“toogle tree”). In the navigation pane (on the left), expand cloned folder `clprog-lab-1-<YOUR-NAME>`.
6. Open file “README.md”. Append your name to the first line. Save file.
7. In the bash (at the bottom) type:

```
$ git status
$ git add .
$ git commit -m "My first commit"
$ git push
```

If you did everything right, you should see that repository is sent to the GitHub.

At this moment make a screenshot of the full screen (in Windows press Ctrl-PrtScr or Win-Shift-S), and save it on disk.

8. Open your assignment on the GitHub, and check if you can see your name in the README.md file.

Once your repository is pushed (stored in the GitHub), you can safely delete your local repository, if you want.

3.2.5 Download SSH keys to your computer

To save your key pair, you can download it to local computer, and store in a secure place.

1. In Cloud9, in files panel (on the left), select gear icon next to “FAVORITES”. Put checkmark next to:
 - Show Home in Favorites
 - Show Hidden Files
2. In files panel, expand `~/.ssh` folder. Right click `<YOUR-NAME>_cloudprog2026` file, and select “Download”.
3. After the lab do **not** delete your Cloud9 instance. You can reuse it next class.

Bibliography

- [1] “Generating a new SSH key and adding it to the ssh-agent.” [Online]. Available: <https://docs.github.com/en/authentication/connecting-to-github-with-ssh/generating-a-new-ssh-key-and-adding-it-to-the-ssh-agent>.
- [2] “Adding a new SSH key to your GitHub account.” [Online]. Available: <https://docs.github.com/en/authentication/connecting-to-github-with-ssh/adding-a-new-ssh-key-to-your-github-account>.
- [3] B. S. Scott Chacon, *ProGit*, 2nd ed. <https://git-scm.com/book/en/v2>; Apress, 2014.